Impact of Marine and Atmospheric Heatwaves on Intertidal Seagrass: Experimental Spectroradiometry and Satellite-Based Insights

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2024-12-09

Seagrasses play a vital role in coastal ecosystems, providing habitat, stabilizing sediments, and contributing to carbon sequestration. However, climate change has increased the frequency and intensity of heatwaves, posing a significant threat to seagrass health. This study investigates the effects of marine and atmospheric heatwaves on the spectral reflectance of the intertidal seagrass Zostera noltei. Laboratory experiments were conducted under controlled heatwave conditions, where hyperspectral reflectance measurements were taken to assess the impacts over time. Heatwaves caused a substantial decline in seagrass reflectance, particularly in the green and near-infrared regions, corresponding to the browning of green leaves. Key vegetation indices, including the Normalized Difference Vegetation Index (NDVI) and Green Leaf Index (GLI), showed pronounced reductions under heatwave stress, with NDVI values decreasing by up to 34% and GLI by 57%. A novel metric, the Seagrass Heat Shock Index (SHSI), was developed to quantify the transition of seagrass leaves from green to brown, demonstrating a strong ability to capture the effects of heatwave exposure on seagrass coloration. These findings highlight the potential of spectral reflectance as a tool for detecting early signs of heatwave-induced stress in seagrasses, offering a valuable method for remote sensing-based habitat assessment. Multispectral satellite observations corroborated the laboratory results, revealing widespread browning of seagrass leaves during marine and

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atmospheric heatwave events in South Brittany, France. Notably, darkened seagrass patches were observed in intertidal areas exposed to temperatures exceeding 32°C for over 13.5 hours per day. This work highlights the need for continuous monitoring of seagrass meadows under the current and future climate regimes, underscoring the potential of remote sensing to capture rapid environmental changes in intertidal zones.

1 Abstract Long version (For LPS Submission)

Seagrasses meadows are important coastal ecosystems, serving as crucial habitats for marine biodiversity, stabilizing sediments to mitigate erosion, and acting as significant carbon sinks in global climate regulation. However, these vital ecosystems face mounting threats from climate change, with the intensification and increased frequency of marine and atmospheric heatwaves posing profound risks to their health and functionality. This study investigates the impact of these extreme thermal events on the intertidal seagrass, employing a combination of laboratory-controlled experiments and satellite-based remote sensing to capture changes in spectral reflectance and assess the broader ecological implications.

During laboratory experiments, seagrass of the species *Zostera noltei* were exposed to controlled simulated heatwave conditions to assess the physiological and structural impacts of extreme thermal stress. The experimental design involved placing seagrass samples in intertidal chambers that simulated natural tidal cycles, allowing to closely regulate temperature conditions during both high and low tides. One chamber served as a control, maintaining typical seasonal temperatures, while the other was used to simulate heatwave conditions with progressively increasing air and water temperatures to mimic an actual heatwave event observed in the field. Hyperspectral reflectance measurements were recorded at regular intervals during low-tide to monitor changes in the seagrass over time. Heatwave exposure resulted in a significant reduction in reflectance, particularly in the green (around 560 nm) and near-infrared (NIR) regions of the spectrum. This decline in reflectance was closely linked to visible leaf browning, suggesting alterations in pigment content, in the internal structure of seagrass leaves and their overall vitality. The green reflectance decline indicates a reduction in the plant's health, while changes in NIR reflectance often relate to the internal arrangement of cells and air spaces, which are sensitive to heat-induced damage. Vegetation indices like the Normalized Difference Vegetation Index (NDVI) and the Green Leaf Index (GLI), which are indicative of the overall health and structural integrity of vegetation, showed marked decreases under heatwave conditions, with NDVI dropping by up to 34% and GLI by 57%. This significant reduction highlights the adverse effects on the seagrass's ability to maintain its normal physiological processes under heat stress. To quantify these changes more effectively, a novel Seagrass Heat Shock Index (SHSI) was developed. The SHSI, applicable on emerged seagrass, was particularly effective in detecting the transition from green leaves to darkened, stressed leaves, providing a sensitive and reliable tool for assessing thermal stress effects on seagrass. By focusing on specific changes in reflectance across certain spectral bands, the SHSI allowed

for a clear differentiation between unimpacted and impacted vegetation, capturing the onset of heat-induced stress with high accuracy. This sensitivity makes the SHSI valuable for early intervention, enabling managers and researchers to identify vulnerable seagrass meadows before substantial damage occurs, thereby facilitating more timely conservation measures.

Complementing the experimental data, Sentinel-2 observations provided clear evidence of the effects of a documented heatwave event in South Brittany, France, on natural seagrass meadows. These intertidal zones were exposed to extreme air temperatures up to 32°C for more than 13.5 hours per day, leading to significant leaf darkening, which affected up to 24% of the meadow's area. The satellite-derived SHSI indicated a strong spatial correlation between prolonged heat exposure and areas experiencing spectral darkening, highlighting the susceptibility of intertidal seagrasses to extended thermal stress. Spatial analysis showed that darkening was especially pronounced in the higher intertidal regions, where seagrasses were exposed to air for longer durations during low tide, underlining the interaction between tidal exposure and thermal stress. However, Sentinel-2 data one month after the heatwave, showed partial recovery in some areas, suggesting a certain level of resilience in *Zostera noltei*. Despite this recovery, the seagrasses that had experienced the most severe darkening did not fully return to their original state, indicating that while seagrass meadows have some capacity for resilience, prolonged or repeated thermal stress can have lasting impacts, particularly in the more exposed intertidal regions. This highlights the need for focused conservation efforts to support their recovery and enhance their resilience in the face of increasing climate-driven thermal extremes.

The combination of laboratory-controlled experiments and satellite-based remote sensing provided a comprehensive understanding of the impacts of heatwaves on seagrass meadows at multiple scales, from individual leaf-level responses to entire meadow-wide effects. This integrated approach highlights the potential of leveraging both detailed local observations and large-scale satellite data to effectively monitor ecosystem changes, offering valuable insights for the management and conservation of these vulnerable habitats. The study underscores the critical role of spectral reflectance as an early warning indicator of heatwave-induced stress, laying the foundation for remote sensing-based monitoring and conservation efforts. By employing innovative indices like the Seagrass Heat Shock Index (SHSI), we capture subtle yet ecologically significant changes, advancing the precision of habitat assessments in intertidal zones. As climate scenarios predict more frequent and intense heatwaves, the need for continuous monitoring of intertidal seagrass meadows becomes increasingly urgent. This research demonstrates the efficacy of remote sensing in capturing rapid environmental changes, providing a framework for mitigating the impacts of climate-driven stressors and calling for adaptive conservation strategies. These strategies should integrate advancements in remote sensing technologies with targeted field-based interventions to preserve the resilience of intertidal seagrass meadows, thereby addressing the escalating challenges posed by climate change and ensuring the continued health of these indispensable coastal habitats.

2 Introduction

Intertidal seagrasses play a crucial role in coastal ecosystems by providing habitats and feeding grounds for various marine species, supporting rich marine biodiversity, and contributing significantly to primary production and carbon sequestration (Sousa et al., 2019; Unsworth et al., 2022). These seagrasses are essential for several ecological functions, such as sediment stabilization or reducing the risk of eutrophication by consuming nutrients (**REF**). This justifies their use as indicators of environmental changes due to their sensitivity to water quality variations (Zoffoli et al., 2021). The interactions between seagrass meadows and their associated herbivores further enhance the delivery of ecosystem services, including coastal protection and fisheries support (Gardner and Finlayson, 2018; Jankowska et al., 2019; Zoffoli et al., 2023). Understanding and preserving these ecosystems is vital for maintaining the biodiversity and productivity of coastal regions (Ramesh and Mohanraju, 2020; Scott et al., 2018).

Despite their crucial role in marine ecosystems, intertidal seagrasses face numerous threats that compromise their health and functionality. Coastal development and human activities are primary threats, reducing the available habitat for seagrasses and increasing water turbidity, limiting light penetration and photosynthesis (Waycott et al., 2009). Seagrasses are also threatened by agricultural and urban runoff nutrient enrichment, leading to eutrophication. This condition promotes the rapid growth of algale causing blooms that compete with seagrasses for light and nutrients, further stressing these important plants (Oiry et al., 2024; Thomsen et al., 2023). Pollution from industrial and agricultural sources introduces harmful chemicals and heavy metals into coastal waters, posing toxic risks to seagrass health (Bastos et al., 2023; Zahoor and Mushtaq, 2023). These pollutants can affect the physiological processes of seagrasses, reducing their growth and survival rates (Sevgi and Leblebici, 2022). Among these anthropogenic stressors, heatwaves, exacerbated by climate change, pose a growing threat to seagrasses (Deguette et al., 2022; Sawall et al., 2021). **However, their impact on seagrass spectral reflectance has not yet been investigated.** Marine Heatwaves (MHW) are defined by Hobday et al. (2016) as prolonged discrete anomalously warm water events while Atmospheric Heatwaves (AHW) are defined by Perkins and Alexander (2013) as periods of at least three consecutive days with temperatures exceeding the 90th percentile of a time series covering at least 30 years. Intertidal seagrasses are exposed to both MHWs and AHWs at the interface between the land and oceans. Heatwaves profoundly impact the physiology of seagrasses, with their effects varying based on species and geographic location. The widespread species *Zostera marina* exhibits high susceptibility to elevated sea surface temperatures during winter and spring, leading to advanced flowering, high mortality rates, and reduced biomass (Sawall et al., 2021). Similarly, *Cymodocea nodosa* shows increased photosynthetic activity during heatwaves but suffers negative effects on photosynthetic performance and leaf biomass during recovery (Deguette et al., 2022). Additionally, different populations of *Zostera marina* along the European thermal gradient exhibit varied photophysiological responses during the recovery phase of heatwaves, indicating differential adaptation capabilities among populations (Winters et al., 2011). These events intensify other stressors, such as overgrazing and seed burial, compromising recruitment (Guerrero-Meseguer et al., 2020). Structural impacts on leaf coloration were also observed: in September 2021, a seagrass meadow in South Brittany (France) exhibited browning of the leaves just a few days after the start of a heatwave characterized by both water and atmospheric temperature anomalies (Barillé A-L & Harin N., pers. comm.).

Remote sensing is increasingly employed to monitor marine ecosystems, including intertidal seagrass meadows (Davies et al., 2024a, 2024b; Oiry et al., 2024; Román et al., 2021). The pigment composition of plants, such as chlorophylls, carotenoids, and anthocyanins, significantly influences their spectral signature in the visible range due to their specific light absorption properties (Davies et al., 2023; Douay et al., 2022; Olmedo-Masat et al., 2020; Ustin and Jacquemoud, 2020). During senescence, the degradation of chlorophyll and the unmasking of accessory pigments result in noticeable changes in leaf coloration and reflectance, including increased reflectance in the red and green wavelengths and shifts in the red-edge position (Boyer et al., 1988; Mariën et al., 2019; Peñuelas et al., 2004). Leaf browning, often observed after stress events, produces reflectance changes similar to those caused by senescence, enabling the detection of vegetation stress through remote sensing (Boyer et al., 1988; Peñuelas et al., 2004). Spectral indices such as the Brown Pigment Index (BPI) and the Photochemical Reflectance Index (PRI) have been developed to assess changes in terrestrial plant physiological status, including oxidative and drought stress (Garbulsky et al., 2011; Skendzic, 2023). While these effects are well-documented in terrestrial plants, the spectral changes associated with senescence and stress in intertidal marine vegetation, subjected to a combination of aquatic and aerial conditions, remain poorly studied.

This study aims to experimentally test the hypothesis that heatwaves alter the reflectance of the intertidal seagrass *Zostera noltei*. Controlled experiments in intertidal chambers were conducted to evaluate the direct impact of heat stress on seagrass reflectance. These findings were then integrated with *in situ* satellite remote sensing data, providing critical insights into the spatial extent and temporal dynamics of heatwave effects on seagrass meadows. By linking experimental results with large-scale observations, the study underscores the potential of remote sensing to enhance our understanding of seagrass responses to extreme thermal events across diverse settings and timescales.

3 Material & Methods

3.1 Laboratory Experiment

3.1.1 Sampling and acclimation of seagrasses

Seagrass samples were taken from a *Zostera noltei* (dwarf eelgrass) meadow Bourgneuf Bay, France (46°57'32.0"N, 2°10'37.0"W) at low tide in June 2024. A metal sampling box was used to sample seagrass from an area of 30 cm by 15 cm and 5 cm deep, maintaining the sediment structure and avoiding damage to the rhizomes and the leaves of the seagrass (Figure 1 A).

This sampling box limited variability between replicates. The sampled sediment, including seagrass, sediment, meiofauna, and macrofauna, was placed in plastic trays. Keeping the entire biota allowed for the natural interactions between components to be maintained and reduced stress on the seagrass. Seawater was added to each tray to avoid hydric stress caused by insufficient moisture during transportation. Simultaneously, seawater was sampled from a nearby site and transported to the lab, where it was filtered using a 0.22 µm nitrocellulose filter to remove all suspended particulate matter. This seawater was used in the acclimation tank and the intertidal chambers. The seagrasses were acclimated for one week with a water temperature of 17°C, matching the *in situ* temperature during sampling, and with light of 150 µmol.s-1.m-2 of PAR photons (Akbar et al., 2020). During high tide, a pump allowed the water to circulate and to re-oxygenate.

Figure 1: Example of the various steps of the experiment. A: Field sampling of seagrass using a sampling coring device; B: Intertidal chamber used during the experiment; C: Seagrass sample inside a chamber during the experiment at high tide; D: Treatment sample at the start of the experiment; E: Treatment sample at the end of the experiment, 2 days after the start of the heatwave.

3.1.2 Experimental design

Two experimental intertidal chambers from ElectricBlue® (Electric Blue, 2023) were used to simulate tidal cycles and control water temperature during high tide and air temperature during low tide (Figure 1 B,C). The chambers were utilized in parallel, with one chamber serving as the control, while the other was used for the experimental treatment.

To closely reproduce field temperatures experienced by seagrasses during the experiment, *in situ* temperature sensors were installed at the sampling site in Bourgneuf Bay in August 2024. Daily maximum temperatures recorded by these *in situ* sensors (T7.3 EnvLoggers from from ElectricBlue®) were compared to measurements from the nearest Meteo France weather station (Annexe A1, Section 7.1). The control chamber was kept at temperatures representing typical seasonal conditions, with water temperatures at 18°C and air temperatures from 19°C to 23°C, following natural daily temperature fluctuations (Figure 2). For the experimental treatment, the air temperature was adjusted to mimic an atmospheric heatwave that affected the seagrass meadow in Quiberon, South Brittany, France (47°35'40.0"N, 3°07'30.0"W), from September 2 to September 6, 2021. On the first day of the experiment, air temperatures in the experimental chamber were set to range from 23°C at night to 35°C during the day, increasing by 1°C daily during three days. Water temperature in the experimental chamber was also adjusted to mimic heatwave conditions, starting at the seasonal baseline (18°C) and rising incrementally by 0.5°C daily to simulate the increasing temperatures during the heatwave. This aimed to reproduce the thermal stress experienced by the seagrass meadow during this heatwave event (Figure 2). The experiment was repeated three times to obtain replicates (hereafter referred to as "Run"). The intertidal chambers provided by ElectricBlue® were equipped with LED lights that emitted low red and infrared radiation. To achieve a Photosynthetically Active Radiation (PAR) intensity of up to 400 $\text{mol} \cdot \text{m}^{2} \cdot \text{s}^{1}$, a filament bulb was added inside the chambers.

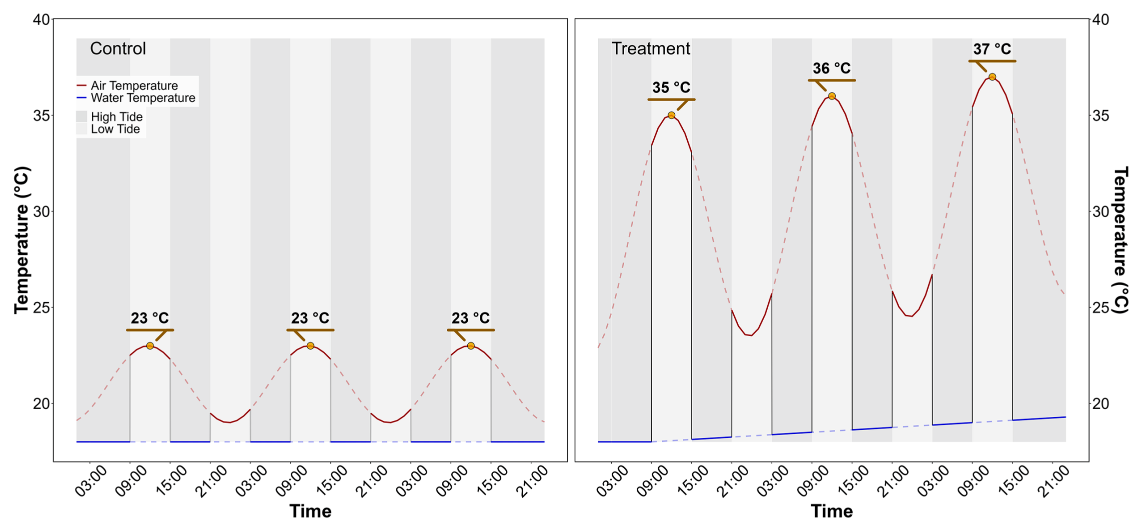

Figure 2: Temperature profiles of both the control (left) and the treatment (right) followed during the heatwave experiment. The red line indicates air temperature, whereas the blue line indicates water temperature. Due to the tidal cycle followed during the experiment, the seagrasses only experience temperatures representade by solid lines.

3.1.3 Optical measurements

3.1.3.1 Hyperspectral measurements

Throughout the experiment, hyperspectral signatures of both the control and treatment seagrasses were taken using an ASD HandHeld 2 equipped with a fiber optic extension, allowing measurements to be taken directly inside the chamber without opening it, where automatic spectra acquisition was carried out (Malvern Panalytical, 2023). An average of five reflectance spectra ($R(\lambda)$), each with an integration time of 544 ms, was taken every minute during daytime. Every 10 minutes, the fiber optic was switched from one intertidal chamber to the other, in order to measure reflectance in both treatment and control. Light conditions were controlled inside of the chambers so reflectance calibration of the instrument was performed each morning at the very first moment of low tide using a Spectralon white reference with 99% Lambertian reflectivity.

3.1.3.2 Spectrum post-processing

A Savitzky-Golay smoothing function with a 5 nm moving window was applied to each spectrum using the "hsdar" package in R (Lehnert et al., 2017). The second derivative at 665 nm (665), showing the highest variability between the control and the treatment, was tested as an indicator of the spectral changes following heatwaves.

Two additional well-established radiometric indices were also tested :

- The Normalized Difference Vegetation Index (NDVI, Rouse et al. (1974)), a proxy of the concentration of chlorophyll-a (Equation 1)

$$NDVI = \frac{R(840) - R(668)}{R(840) + R(668)} \tag{1}$$

where $R(840)$ and $R(668)$ are the reflectance at 840 nm and 668 nm respectively.

- The Green Leaf Index (GLI, Louhaichi et al. (2001)), a measurement of the greenness of seagrass leaves (Equation 2)

$$GLI = \frac{[R(550) - R(668)] + [R(550) - R(450)]}{(2 \times R(550)) + R(668) + R(450)} \tag{2}$$

where $R(550)$ and $R(450)$ are the reflectance in green at 550 nm and in the blue at 450 nm, respectively.

Based on the observed spectral changes in seagrasses exposed to heatwaves, we developed a new radiometric index to detect colour changes in the visible but also near-infrared slope variations, a significant spectral feature for seagrasses (Davies et al., 2023). The observed green colour of the leaves turning brown was characterized by a reduction in reflectance within the green region and a substantial decrease in the red-edge portion of the electromagnetic spectrum.

the Seagrass Heat Shock Index (SHSI) is introduced here to measure the difference between the interpolated value between 560 and 842 nm and the observed value at 740 nm (Figure 3). This approach calculates the line height using a triplet of bands, where 560 nm and 842 nm serve as the 'baseline' and 740 nm represents the central peak. The SDI can be mathematically expressed as:

$$\text{SHSI} = I_{SHSI} - R(740) \tag{3}$$

where :

$$I_{SHSI} = R(560) + \tau[R(842) - R(560)]$$

and :

$$\tau = \frac{740 - 560}{842 - 560}$$

Where $R(560)$, $R(740)$, and $R(842)$ represent reflectance values at 560, 740, and 842 nm, respectively. These wavelengths were selected to align with the spectral resolution of satellites like Sentinel-2, enabling the index map leaves browning from satellite data. When applied to unimpacted seagrass spectra, the interpolated line between 560 and 842 nm falls below the reflectance value at 740 nm, resulting in a negative SHSI value (Figure 3 A). In contrast, when applied to impacted seagrass spectra, the interpolated line lies above the 740 nm reflectance value, resulting in a positive SHSI value (Figure 3 B).

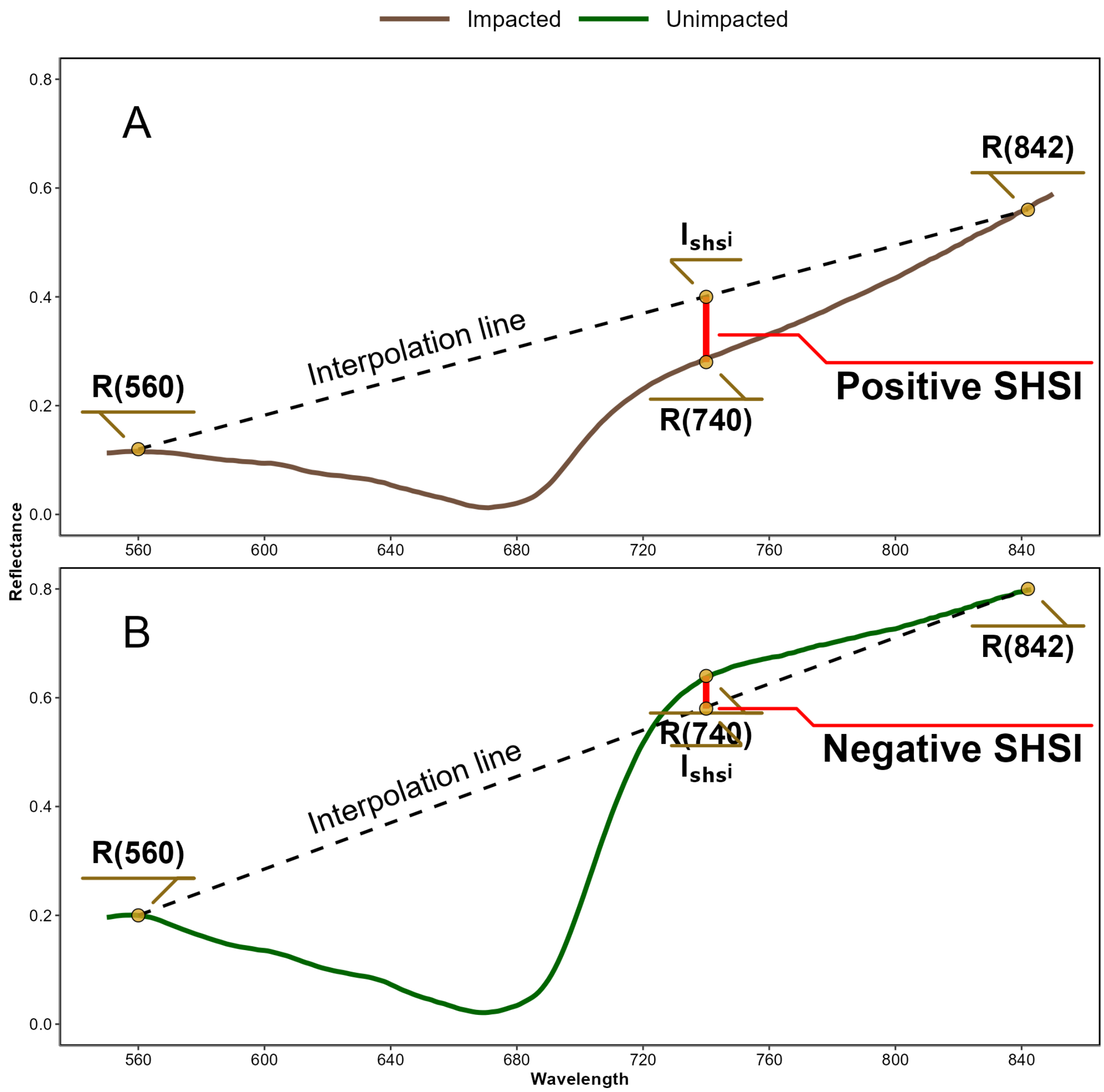

Figure 3: Computation of the Seagrass Heat Shock Index (SHSI) from Reflectance Spectra of Impacted (A) and Unimpacted (B) seagrass. Hard darkline represents the interpolation between 560 and 842 nm while the red line represents the SHSI value, mesured at 740 nm.

3.2 Observation of a seagrass bed impacted by a heatwave

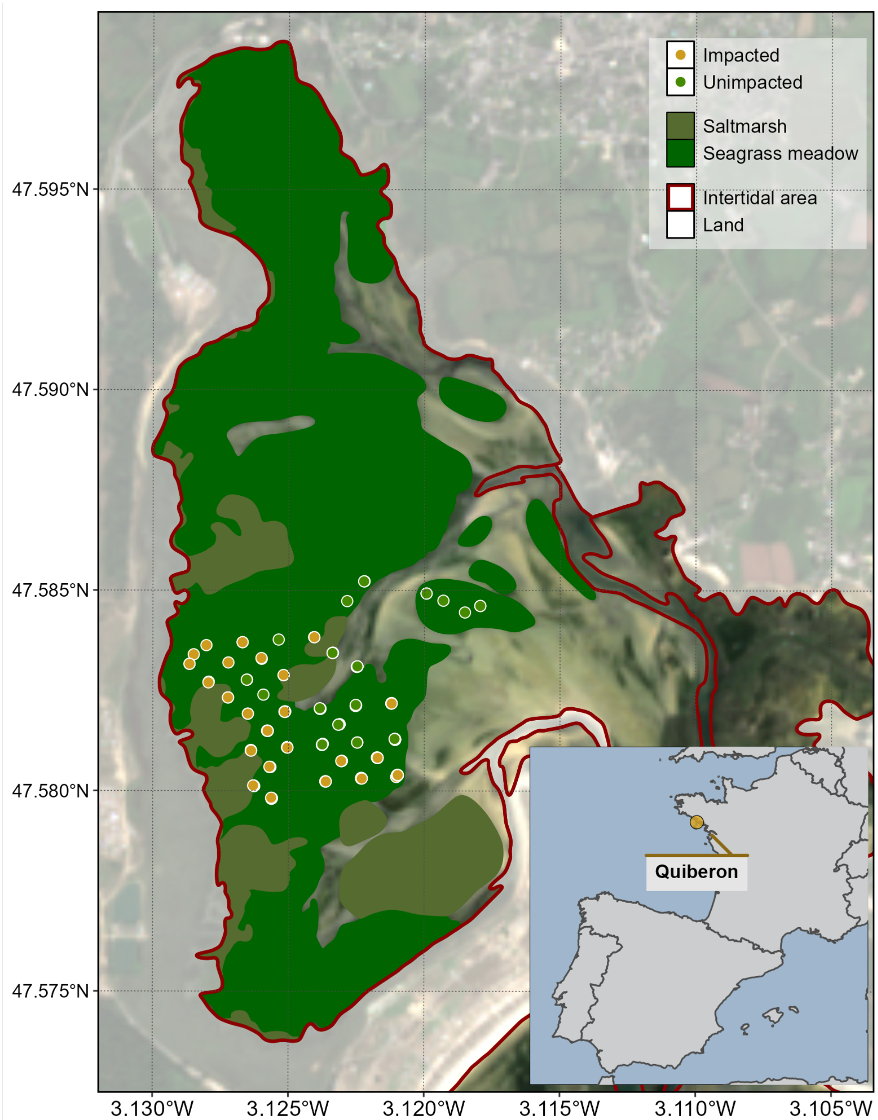

Figure 4: Location of field observation in a seagrass meadow impacted by a heatwave that occured on the 10th of September 2021 in Quiberon, South Brittany, France. The red line indicates the intertidal zone (Zone between high tide and low tide, exposed during low tide), white polygons indicates land areas, the dark green polygon indicates the extent of the seagrass meadow and the olive polygons indicates saltmarshes. Greendots indicate the location of quadrat picture over unimpacted seagrasses (i.e. showing a green colour on the field), while orange-red dots indicate the location of the quadrat taken over impacted seagrasses (i.e. showing a brown colour on the field).

A field campaign, aiming to map a seagrass meadow near Quiberon (France : 46°57’32.0”N, 2°10’37.0”W), occurred on September 10th 2021 after a heatwave (Figure 4). Seagrass leaves turning brown were observed over large areas of the meadow alongside normal green seagrass (Figure 5). A total of 96 Quadrat Points (QPs) were collected as georeferenced quadrat images across the meadow. These images allow for visual assessment of vegetation type, density, and coloration. The quadrats were then divided into two categories: green seagrasses (henceforth: QPs unimpacted) and brown seagrasses (henceforth: QPs impacted), based on a visual estimation of the leaf coloration (Figure 4).

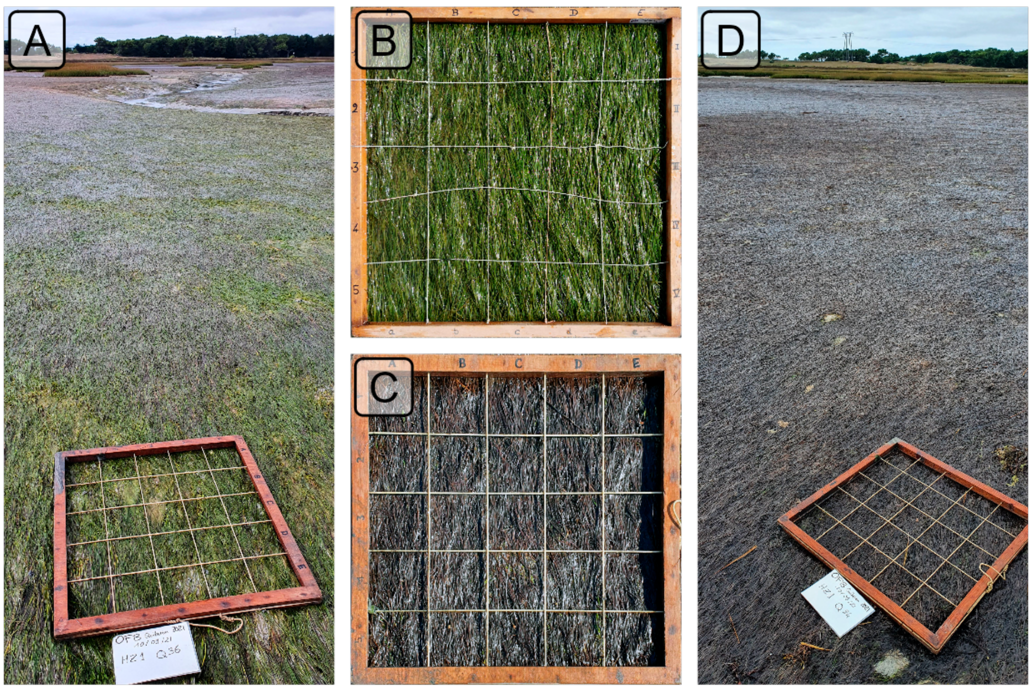

Figure 5: Illustrations of the two colorations of seagrass leaves observed in the field. A: View of a green meadow; B: Quadrat with green leaves; C: Quadrat with brown leaves; D: View of a meadow with all leaves turned brown. All images were taken on September 10th, 2021, in Quiberon, South Brittany, France, after a heatwave.

3.2.1 Temperature data and identification of the heatwave

3.2.1.1 Air temperature

Since January 1st, 2024, Meteo France weather data has been freely and openly accessible. Hourly air temperature data (°C) was retrieved using a custom script as no API was available at the time of this study. Data from the nearest weather station of our study site (Lorient-Lann Bihoue, 47°45'46"N 3°26'11"W) were downloaded. For this station more than 395000 observations since the February 1, 1952, were available.

3.2.1.2 Water temperature

SST data from the Copernicus CMEMS platform were downloaded for the French coast near Quiberon (47°29 03 N, 3°07 09 W), covering 1982-2022 (CMEMS, 2024). Only pixels within an area of 2700 km² around Quiberon, Brittany, France (47°29 03 N, 3°07 09 W) were extracted and analyzed. This area was large enough to minimize missing values caused by cloud cover yet small enough to avoid being influenced by the stability of offshore SST.

3.2.1.3 Heatwave detection and characterization

Marine and Atmospheric Heatwave detection was performed using the HeatwaveR package in R (Schlegel and Smit, 2018). This package utilizes the methodology proposed by Hobday et al. (2016) to detect heatwave events. The climatology of both air and water temperature for the year was computed using the temperature time series. An event was considered a heatwave each time the temperature exceeded the 90th percentile of the climatology for three consecutive days. The severity of each event has been assessed using the methodology proposed by Hobday et al. (2018).

3.2.2 Satellite observation

Three Sentinel-2 images covering the field campaign site (Figure 4) were downloaded from the Copernicus platform (European Space Agency, 2024a). The first image was taken before the heatwave event on September 1st, 2021, the second image during the event on September 6th, 2021 and the last one, one month after the events on the 8th of October 2021. Images were acquired as Level-2 product, in surface reflectance, and were orthorectified and atmospherically corrected to account for the effects of the atmosphere on reflectance values (European Space Agency, 2024b).

The reflectance value of each QPs (Figure 4) was extracted and compared before and after the heatwave event.

3.2.3 Emersion time of the seagrass meadow

To estimate the spatial distribution of emersion time of the seagrass meadow during low tide, bathymetric and tidal data were used. The hypothesis was that the leaf's discoloration follwing the heatwave was related to the emersion time and intertidal topography. Bathymetry data for the Quiberon intertidal meadow were sourced from the "Service Hydrographique et

Océanographique de la Marine" (SHOM, 2021). It corresponds to the Litto3D® product, with a spatial resolution of 1 m per pixel, which is a three-dimensional land-sea elevation database, accurately depicting the coastal terrain along certain regions of the French shores. It uses the NGF/IGN69 reference "zero", which corresponds to the mean sea level recorded at the Marseille tide gauge between 1885 and 1897, commonly known as the "Terrestrial Altimetric Zero".

Tidal data during the heatwave event were downloaded from the Intergovernmental Oceanographic Commission data portal (IOC, 2024), using measurements from the nearest tide gauge at Le Crouesty. In this dataset, the reference "zero" corresponds to the lowest astronomical tide, also known as the Hydrographic Zero.

Before calculating the emersion time, both datasets were intercalibrated to a common altitude reference. This involved applying a correction factor to the Litto3D data to align it with the Hydrographic zero. SHOM annually publishes a document called "Références Altimétriques Marines" (RAM, Shom, 2022), which provides these correction factors for each seaport along the French coastline. The correction factor for Le Crouesty port data from 2022 was 2.85 m.

Once aligned, the corrected elevation was compared to water height for each pixel and each time step during the heatwave event. The emersion time was then calculated as the daily total time each pixel remained exposed.

3.3 Statistics

General Linear Models (GLMs) were used to assess relative differences over time in response variables (Spectral Indices) with different treatments. For analysis of relative changes in indices calculated from experimental spectra the relative change in index was modelled as function of Days (1-3: Discrete) with Replicate (henceforth Runs; 1-3: Factor) and Timestep within Run (1-6: Factor) as cross random factors. Satellite derived indices were modelled as function of Date (1-3: Discrete) and Treatment (Impacted vs Unimpacted: Categorical). A General Additive Model (GAM) was used to assess the relationship between relative SDI change with emersion time. SDI was modelled as function of emersion time with a basis spline. All model parameters were estimated with a Bayesian framework using the "brms" and "RStan" packages in R to leverage the stan language (Bürkner, 2021; Carpenter et al., 2017; R Core Team, 2023; Stan Development Team et al., 2020). The response variables were modelled assuming a Gaussian distribution, with weakly informative priors (Student-T(3,0,2.5)). Model parameters were estimated using Markov Chain Monte Carlo (MCMC) sampling, with 4 chains of 5000 iterations and a warm-up of 500.

4 Results

4.1 Laboratory Experiment

4.1.1 Seagrass spectra variations related to a heatwave

In the Control group, reflectance remained relatively stable over time, with only minimal changes in magnitude and spectral features (Figure 6). Overall, the Control group's reflectance spectra displayed a peak at 560 nm (green region), low values at 665 nm (indicative of strong chlorophyll-a absorption), and a high plateau in the near-infrared (NIR), beyond 705 nm (Figure 6, left).

In contrast, the Treatment group simulted heatwave showed more pronounced changes in reflectance throughout the experiment (Figure 6, right). At the beginning (Day 1), reflectance values were comparable to those in the Control group, especially in the visible region, with a notable peak around 560 nm and a pronounced trough at 665 nm. However, starting from day 2, reflectance began to decrease across all wavelengths, particularly around 560 nm and in the NIR. At day 3, the NIR reflectance appeared to stabilize at values like those observed during day 2. In the green region, however, reflectance continued to decline slightly until the experiment's conclusion.

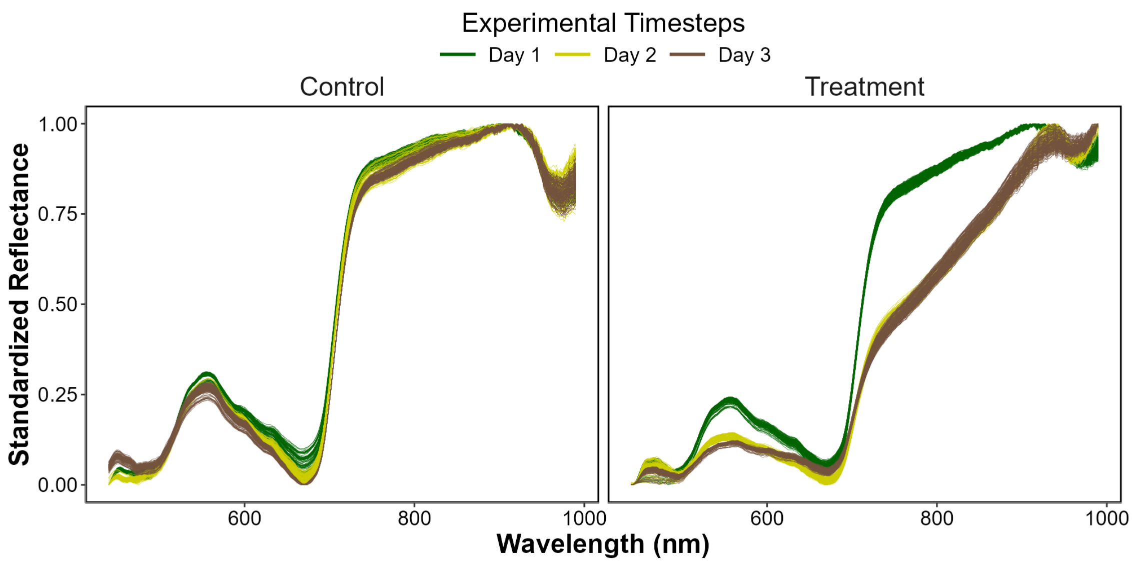

Figure 6: Standardized hyperspectral reflectance signature of *Z. noltei* leaves measured during the heatwave experiment for both the Control (Left) and the Treatment (Right). Color denotes the progression along the experiment from the beginning (Day 1: green), middle (Day 2: Yellow) and end (Day 3: Brown). A min-max standardization has been applied to each individual spectrum.

4.1.2 Evolution of spectral metrics

Similar patterns were found for the ($R''_{665\,\mathrm{nm}}$), NDVI and GLI, where the indices start similar to the control and consistently decline over the 3 days of experiment (Figure 7 A, B and C)

At the start of the experiment (day 1) $R''_{665\,\mathrm{nm}}$, NDVI and GLI in the Treatment group were on average 13% greater, 3% lower and 2% lower, respectively, than that of the Control. However, by day 2, the Treatment showed a rapid decease of approximately 27%, 17% and 28%, eventually reaching a total decline of 68%, 31% and 54% by day 3.

Unlike the other metrics, at the start of the experiment (day 1), the SHSI for the Treatment group was on average 55 % greater than that of the Control. By day 2, the SDI for the Treatment exhibited a rapid decrease of approximately 241 %, eventually reaching a cumulative decline of 420 % by day 3 (Figure 7 D).

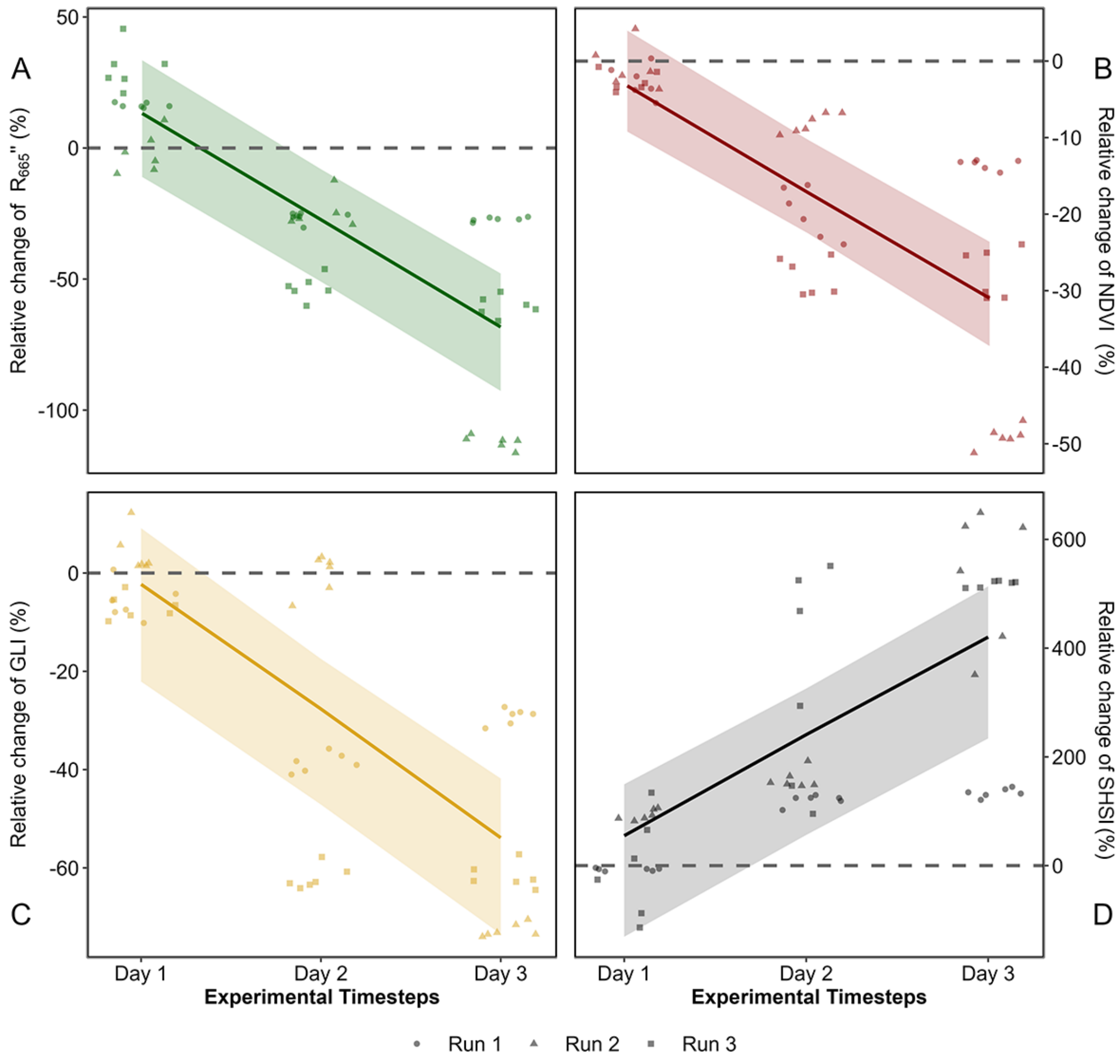

Figure 7: Comparison of spectral metrics for detecting reflectance changes of seagrass leaves after a heatwave. A: Relative difference of the second derivative at 665 nm between the Control and the Treatment over time; B: Relative difference of the NDVI; C: Relative difference of the GLI; E: Relative difference of the SHSI. Points indicate raw data, the line represents a generalized linear model, and the shaded area is the model's standard error. The dased lines represent no difference between the Control and the Treatment.

The SHSI revealed clear distinctions between the Control and Treatment groups (Figure 8). At day 1, the SHSI of the Control and Treatment groups are comparable, with value of -0.11 and

-0.08, respectively. By the end of the experiment, seagrasses in the Treatment group exhibited an average SDI of 0.15, consistent with their visibly darkened appearance. In contrast, the Control group retained a green appearance throughout the experiment, with an average SDI of -0.07.

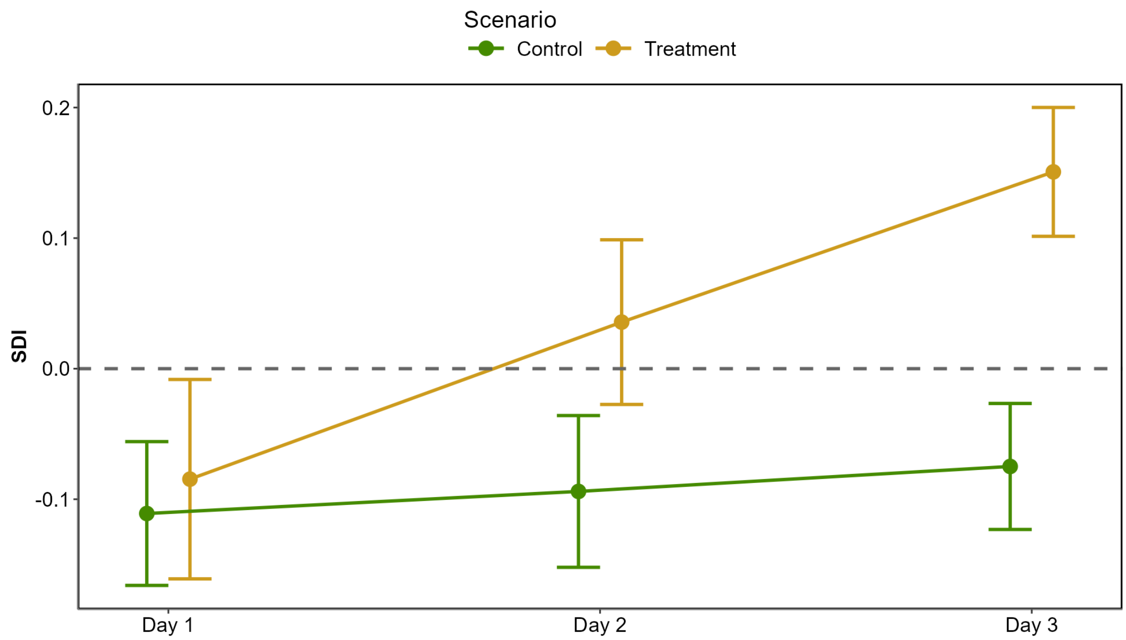

Figure 8: Average and standard deviation of the Seagrass Darkening Index (SDI) across expimental runs, at each day of the experiment. Tge green line shows values of the Control group while the golden line indicates values of the Treatment group.

4.2 Heatwaves of Quiberon

4.2.1 Spectral inspection

The Sentinel-2 images used in this study are dated from the 1st of September 2021 and the 6th of September 2021 (Figure 9 A and C, respectively). The Atmospheric Heat Wave (AHW) started on 4 September and lasted until 7 September, while the Marine Heat Wave (MHW) started on 3 September and ended on 8 September 2021 (Figure 9 B). The AHW has been classified as a strong event, and the MHW was classified as moderate. Even though the Sentinel-2 of the 6th of September was captured only a few days after the start of the events, darkening of seagrass can already be observed in the true-color composition (Figure 9 C). Before the event, all QPs appeared green on the images, with similar reflectance spectra, regardless of the class attributed to the QPs in the field after the events (Figure 9 A and D left). Their reflectance spectra showed a peak at 560 nm (in the green part of the spectra), low values at 665 nm and a high infrared plateau (> 705 nm). However, on the 6th of September, QPs classified as impacted during the field campaign, showed significant differences in reflectance spectral shape compared to QPs classified as unimpacted during the campaign, with corresponding differences in the true color composition (Figure 9 C and D right). The reflectance spectra over dark seagrass were characterized by the loss of the reflectance peak at 560 nm and a decrease in the infrared plateau, which was observed as a steady increasing slope up to 940 nm.

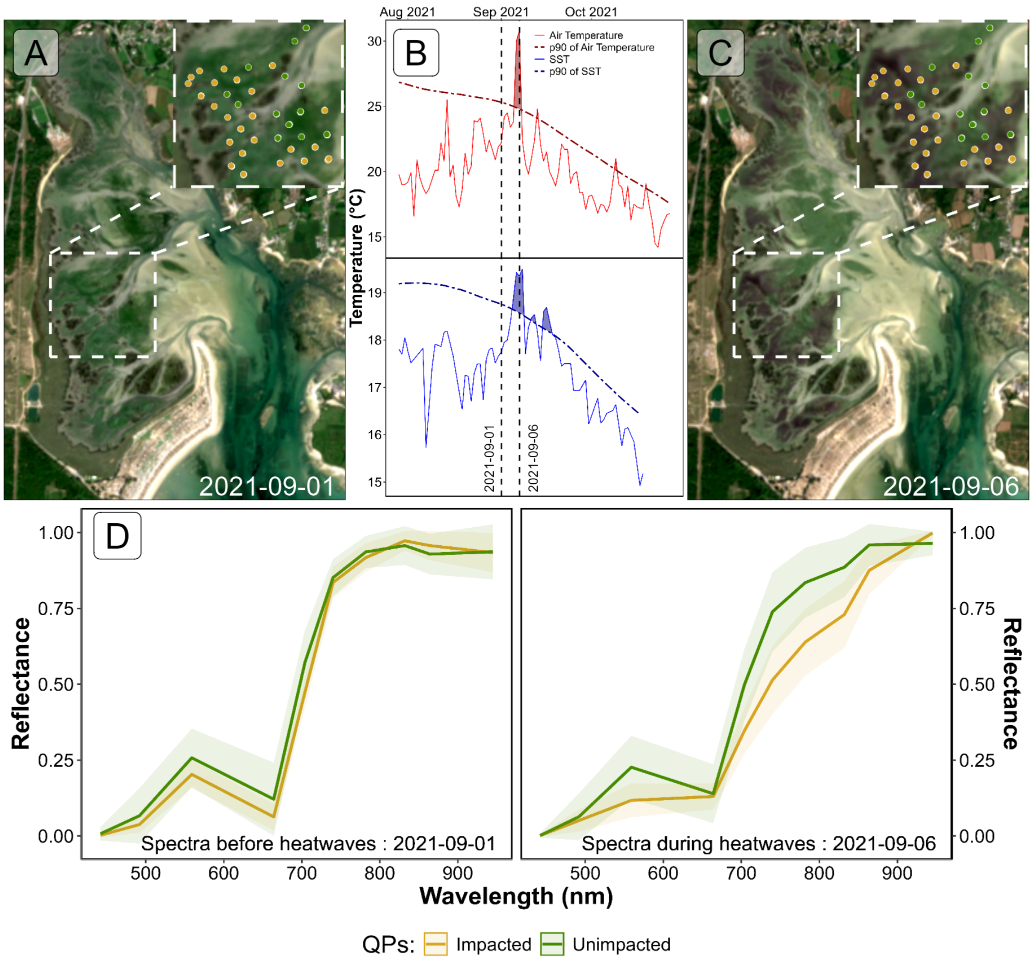

Figure 9: Spectral Comparison of Sentinel-2 Reflectance (D) of Quadrats Points (QPs) Before (A) and During (C) Heatwave Events (B); A: RGB color composition of the Sentinel-2 image on the 1st of September 2021. The points correspond to QPs collected on September 10, 2021, with unimpacted seagrass in green and impacted seagrass in golden; B: Detection of heatwave events based on both Air Temperature and Sea Surface Temperature (SST). The solid line represents the daily average temperature, while the dashed line indicates the 90th percentile of the climatology. Colored areas identify heatwaves (marine in blue and atmospherical in red). The two vertical dashed lines represent the acquisition dates of the two Sentinel-2 images used in this analysis (01-09-2021 and 06-09-2021) ; C : RGB color composition of the Sentinel-2 image on the 6th of September 2021, with unimpacted seagrass in green and impacted seagrass in golden; D : Average spectral signatures of QPs where dark and green seagrasses (gold and green lines, respectively) were identified during the field survey. The left plot shows the reflectance of these QPs before the heatwave impact, while the right plot shows the spectral signature during the heatwaves. The ribbons around the lines represent the standard deviation.

4.2.2 Sentinel-2 derived metrics comparison

Using Sentinel-2 data, NDVI, Seagrass Cover (SC), GLI and SDI estimated for pixels unaffected by the heatwave showed minimal changes for all metrics of -1 %, -1 %, -10 and 3 %, respectively between the 1st of September and the 6th of September (Figure 10). In contrast, seagrass impacted by the heatwave exhibited a significant changes in all metrics. Because the SC is linearly related to the NDVI, both metrics dropped of 35 %, however the drop in SC was not due to a loss of density of seagrass on the field (Figure 5). The GLI drops of about 85% at the 6th of September. The SDI was the most sensitive metric to the darkening of seagrasses, showing an increase of 97 % during heatwave exposure (Figure 10).

One month later the heatwaves events, on the 8th of October, all metrics of unimpacted QPs had increased compared to the 1st of September: The NDVI and SC by 21 %, The GLI by 15 % and the SDI 14 %. Concerning QPs impacted by the heatwaves, even though all the metrics were closer to the values they had prior to the heatwaves, they have not regained their level from before the events. By the 8th of October, the NDVI and the SC of these QPs were 5 % lower, The GLI was 16 % lower and the SDI 14 % higher than on the 1st of September.

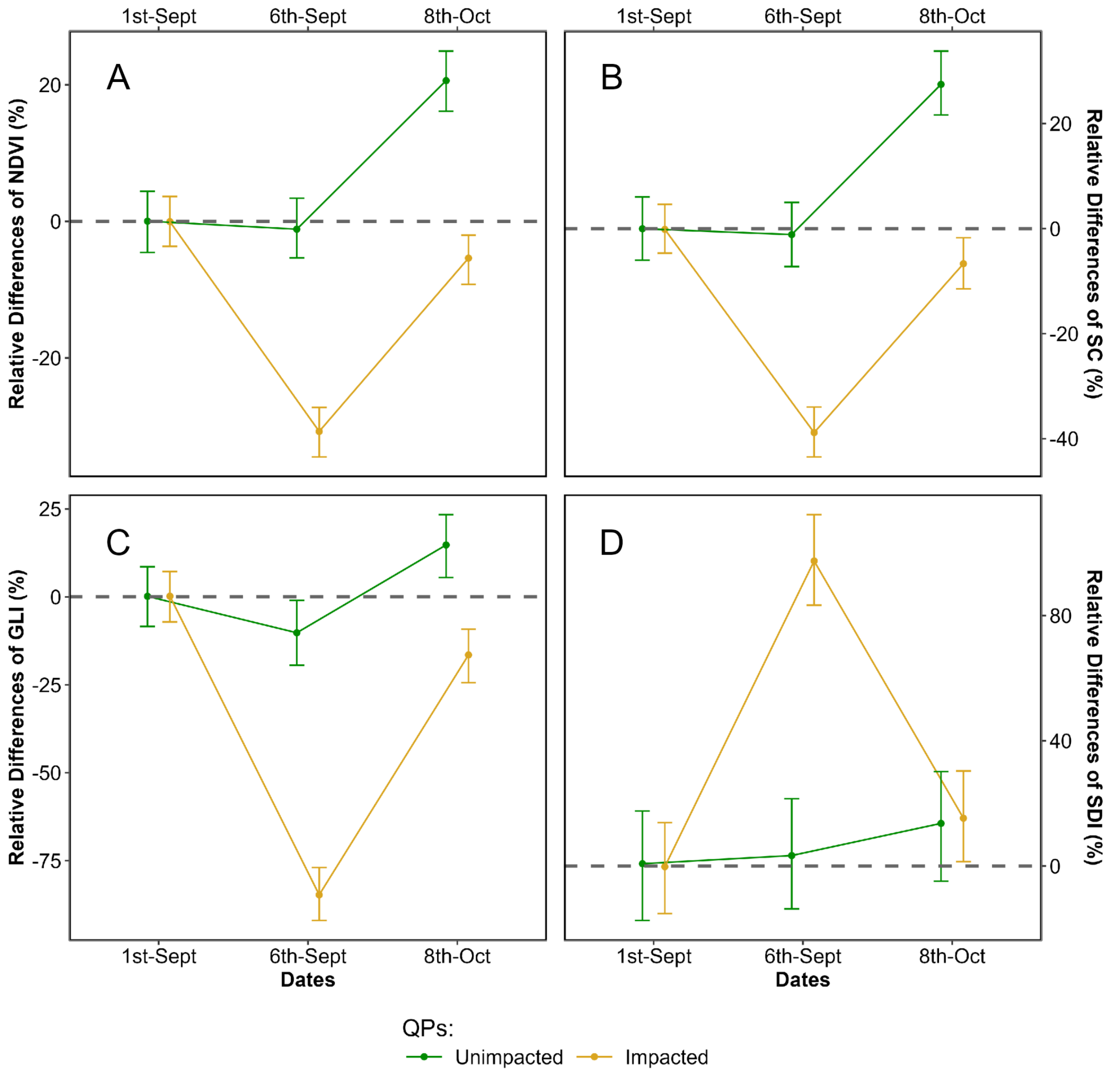

Figure 10: Relative difference of the Normalized Difference Vegetation Index (NDVI: A); Seagrass Cover (SC: B); Green Leaf Index (GLI: C); and Seagrass Darkening Index (SDI: D) to the 1st of September 2021 of Sentinel-2 based metrics at two different dates : during (2021-09-06) and after the heatwave events (2021-10-08) of Quiberons, for the two category of QPs : Unimpacted seagrasses (green) and Impacted Seagrasses (golden). Points represent predicted value of the metric using a Generalized Linear Model (GLM) while error bar represent the 89% confidance interval of it.

4.2.3 Mapping of the seagrass darkening

Using the SDI (Equation 3), we detected seagrass patches that have darkened (Figure 11). Following the heatwave events, a total of 26.9 hectares of seagrass darkened, with the largest affected patch covering nearly 8 hectares. Overall, 24 % of the total seagrass meadow area showed signs of darkening between the 1st and 6th of September 2021. Comparing the spatial distribution of darkened patches with the site's bathymetry revealed that 94.6 % of darkened areas were located at above 3.9 meters of bathymetry (Figure 11, A and B). One month later, on October 8th, areas that were previously darkened had regained their green color (Figure 11, C).

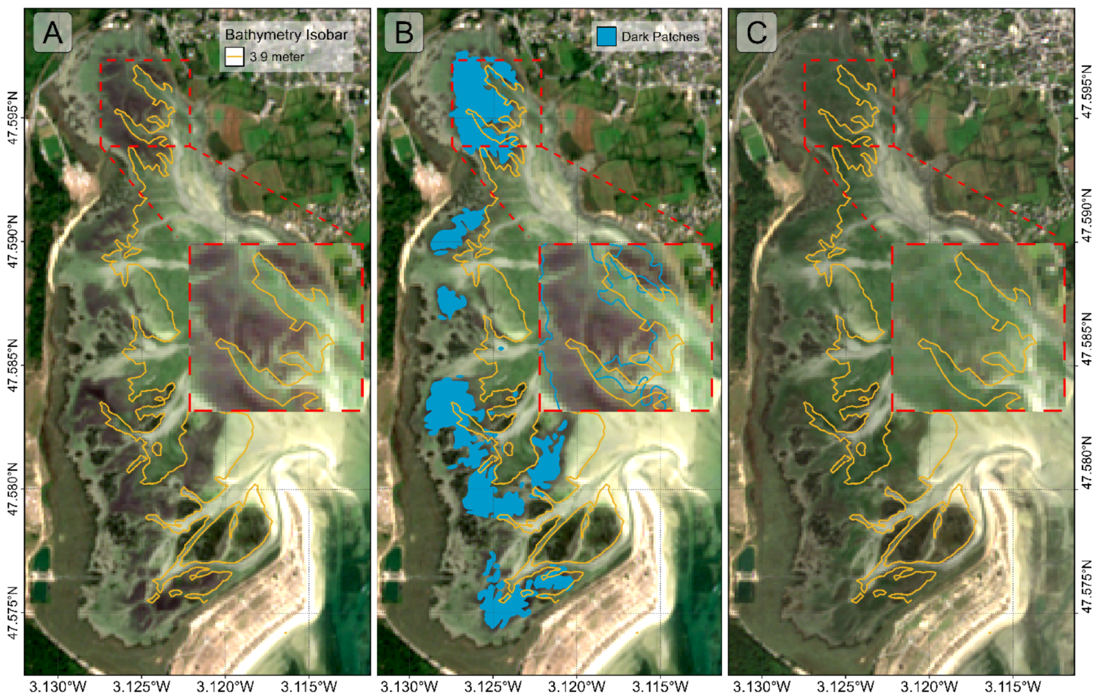

Figure 11: Map of the Darkening of seagrasses during the heatwave events of September 2021 in Quiberon. A: 3.9 m bathymetry isobar (golden line); B: Result of the detection of darkened seagrass patches (Blue polygons); C: Sentinel-2 image one month after the end of the heatwaves (2021-10-08).

Additionally, seagrass emersion time revealed a clear relationship between the duration of air exposure and seagrass darkening (Figure 12). During the heatwave, no significant darkening occurred with daily exposure less than 13 hours. However, above 13 hours, seagrasses began to darken, reaching peak darkening at around 14.5 hours of daily exposure.

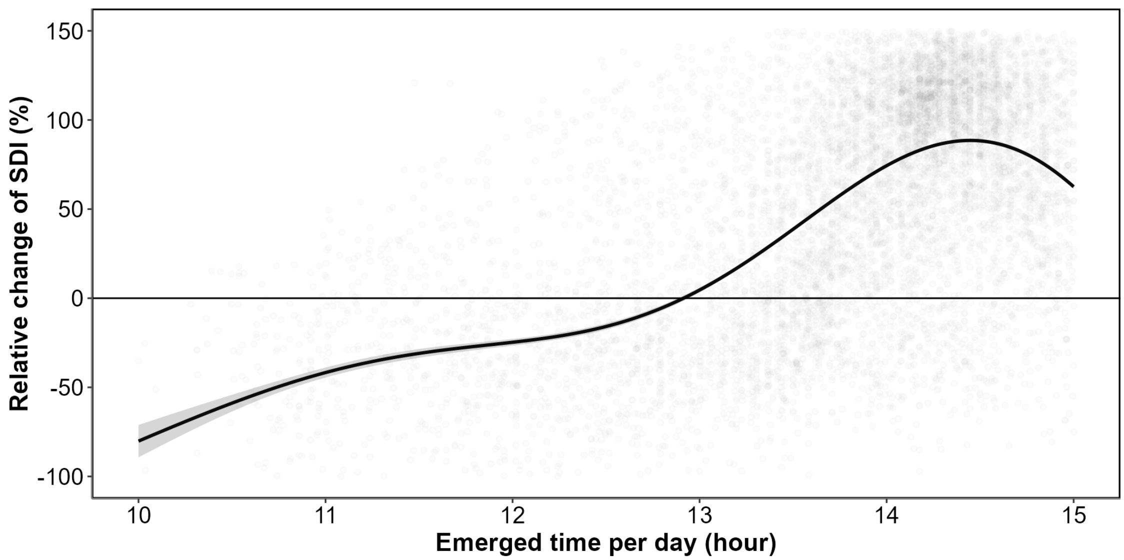

Figure 12: Relative change of the Seagrass Darkening Index (SDI) before and during the heat-waves events as a function of the daily emergence time of seagrasses. The line represents a Generalized Additive Model (GAM) prediction, and the shaded area indicates the standard error. Shaded points represent raw data, with each point corresponding to a single pixel of the meadow.

5 Discussion

5.1 Physiological impacts of heatwaves on seagrasses

Although extensive research exists on marine heatwaves' effects on subtidal seagrasses, little attention has been given to intertidal habitats and even less to the interaction between atmospherical extreme event and intertidal meadows. This study initiates an exploration of how intertidal seagrasses respond to the dual influence of marine and atmospheric heatwaves, underscoring the need for further investigation in this underexplored area.

In the present study, significant changes in the reflectance of seagrasses exposed to heatwaves were observed experimentally. These changes mainly include a drop in reflectance around 560 nm and another drop around 740 nm in the near-infrared part of the spectrum (Figure 7). The second derivative at 665 nm (R' '665), NDVI, and GLI (Figure 7 B, C, and D) all demonstrated a clear decline over the experimental period (days 1-3), suggesting a progressive reduction in photosynthetic activity or pigment levels in the treatment group relative to the control. In contrast, SDI shows a marked increase, reaching up to 600% for some samples by day 3 (Figure 7 E). This positive trend in SDI aligns with its design as an index to indicate darkening of seagrass, likely resulting from structural or physiological changes, such as degraded pigmentation or altered light absorption. While the general trends are consistent across experimental runs, there is some variability, particularly evident in the confidence intervals, which are wider for SDI than for the other indices. This suggests that while darkening is a consistent response, it may vary in intensity between individual samples or experimental runs. Furthermore, the contrasting magnitudes of change—especially the pronounced increases in SDI versus the declines in other indices—highlight the sensitivity of SDI to changes associated with seagrass darkening, which could be indicative of stress or adaptation responses specific to the treatment conditions.

The change in color can be multifactorial and has been documented in plants under various stress conditions, including thermal stress (Dascaliuc et al., 2007; Jones and Clayton-Greene, 1992). Leaf blackening in angiosperms, as observed in *Protea neriifolia*, is primarily driven by oxidative processes involving the enzyme polyphenol oxidase (PPO) and the oxidation of phenolic compounds. When photosynthesis is inhibited by factors such as low light, chemical interference, or thermal stress, the plant's production of essential carbohydrates and antioxidants diminishes, increasing oxidative stress and leading to darkening. Experimental essays have proved that low-oxygen environments and the addition of antioxydants like diphenylamine (DPA), have been effective in reducing these oxidative reactions. In the absence of photosynthesis, membrane integrity is also compromised, allowing PPO to interact with phenolic compounds, thereby accelerating darkening. Furthermore, high temperatures can destabilize membranes, especially in chloroplasts, disrupting photosystem II and impairing recovery of photosynthetic function. As chlorophyll-a degrades, the ratio of pigments shift; with pigments like carotenoids becoming more prominent, leading to a darkening of the leaves.

Zostera noltei, as a species inhabiting the intertidal zone and regularly exposed to air, has developed adaptations to minimize water stress. For example, it exhibits smaller, narrower leaves compared to species residing lower in the intertidal zone, such as *Zostera marina*, which helps reduce water loss during air exposure periods (Cabaço et al., 2009). However, during intense warming events and under high light conditions, desiccation can occur in certain parts of the meadow, particularly where seagrass leaves are exposed for prolonged periods (Figure 12). Under such water stress, cellular turgor pressure decreases (the internal cell water pressure that maintains cell shape), and concentrations of ions like $Na+$ and $Cl-$ can reach toxic levels. These effects can impair cellular functions, enzymatic activity, and membrane stability.

High infrared reflectance in plant leaves is primarily due to the internal cellular structure that scatters light. Once sunlight enters the leaf, it is diffused through carious layers, particularly the spongy mesophyll, which contains air spaces and cell walls with different refractive indices. this scattering effect is intensified because there is minimal absorption in the near-infrared range (700 - 1300 nm), allowing light to reflect back through the leaf surface. Once the membranes (chloroplasts, thylakoids, cell walls) are destroyed due to oxidative stress, the reflectance in the Red Edge and the near-infrared regions of the spectrum ($>$ 700 nm) decreases significantly (Knipling, 1970).

These unique reflectance properties of seagrasses under heat stress enable the detection of this darkening through remote sensing techniques. The SDI (Equation 3) index developed in this study leverages key reflectance bands (560, 740 and 840 nm) to detect reductions in the green and the Red-Edge regions. SDI can be used to assess the extent and severity of darkening events across intertidal seagrass meadows from space. Current satellite missions, including Sentinel-2, Pleiades-Neo, WorldView-3, SkySat, and GeoSat-2 along with upcoming missions (Sentinel-2 Next Generation and Landsat Next), capture reflectance in the three wavelengths used as input for SDI. As climate change advances, remote sensing becomes a crucial tool for monitoring cumulative ecological impacts on seagrass meadows due to heatwaves. Although these physiological and structural changes occur at the cellular level over short temporal scales, they manifest synchronously across the entire meadow. Remote sensing, with its synoptic views and real-time acquisition capabilities, enables the monitoring of these rapid biological responses to natural stressors. However, cloud coverage can limit continuous observations, necessitating the integration of alternative tools, such as multi-spectral drone platforms. Combined approaches allow for precise, localized monitoring of damaged seagrass meadows immediately after extreme events, which is critical for the early detection of ecosystem degradation, quick actions for habitat conservation and targeted management strategies. Moreover, uninterrupted, multi-year data acquisition would strengthen predictive ecological models for future heatwave impacts, enhancing the capacity for adaptive management and supporting long-term resilience planning for seagrass ecosystems.

5.2 Climate change and heatwaves

The rapid global escalation of heatwave frequency, intensity and duration is a defining characteristic of the current climate crisis, heavily influenced by anthropogenic activities and greenhouse gas emissions (Devi et al., 2024; Russo and Domeisen, 2023). Recent research suggests that the magnitude of these events has surged in the recent decades, with climate projections predicting a continuation of this trend. For example, heatwaves that were once considered rare are now up to ten times more likely to occur, with some regions experiencing events three times as intense as those in the early 20th century. Cumulative indices that quantify heatwave intensity are based on total temperature exceedances and offer a more comprehensive understanding of event severity than average temperature alone. This is because the cumulative impact of prolonged high temperatures imposes more extensive stress on ecosystems than isolated peak temperatures (Russo and Domeisen, 2023).

From a hazard analysis perspective, the concurrent evaluation of intensity, frequency, and duration of atmospheric heatwaves allows for a partial understanding of their likely impacts. Models like the Heatwave Intensity Duration Frequency (HIDF) provide insights into various heatwave scenarios, indicating that both short intense and prolonged moderate events present unique risks depending on the region. For instance, in the Mediterranean, the frequency of high-intensity heatwaves has dramatically increased, leading to severe impacts on urban infrastructure, public health, energy consumption but also to terrestrial and intertidal ecosystems, such as seagrass meadows (Mazdiyasni et al., 2019; Smale et al., 2019). Sharper increase in extreme heat events are expected in regions experiencing large daily thermic amplitude, which underscores the role of atmospheric variability in local heatwave dynamics. In mid-latitudes, where daily variability is expected to decrease, heat extremes may stabilize at elevated levels, reducing cold extremes while allowing for increasingly frequent hot day (Simolo and Corti, 2022).

In the ocean, similar trends are observed with marine heatwaves (MHWs). MHWs have increased by over 50% in annual days since the early 20th century, and projections suggest that a near-permanent state of marine heatwaves, based on nowadays baselines, could develop by the end of the century if greenhouse gas emissions remain high (Oliver et al., 2019). Events like the "Blob" in the northeast Pacific (2013–2016) have underscored the ecological ramifications of such heat anomalies, including shifts in species distributions, mass mortalities, and habitat degradation across vast oceanic regions. The physical mechanisms driving MHWs, such as altered ocean circulation and air-sea heat flux, further illustrate the interconnectedness of atmospheric and marine systems in the context of climate-driven thermal extremes (Smale et al., 2019).

The escalating frequency and intensity of heatwaves represent not only an atmospheric anomaly but a profound disruption to ecological stability across diverse ecosystems (Devi et al., 2024; Stillman, 2019). The sustained high temperatures associated with both atmospheric and marine heatwaves lead to physiological stress, habitat degradation, and increased mortality in many species, particularly those with limited thermal tolerance (Oliver et al., 2019; Simolo and Corti, 2022). Terrestrial and marine ecosystems experience shifts in species distributions, altered community dynamics, and reduced biodiversity as species are either forced to migrate or face local extinction under increasingly inhospitable conditions (Pansch et al., 2018). Similarly, in marine environments, prolonged heat stress from marine heatwaves has cascading effects on foundational species, including corals, kelps, and seagrasses, all of which are crucial for providing habitat, food, and shelter to diverse marine life (Oliver et al., 2019; Smale et al., 2019). Seagrasses, in particular, play a vital role in carbon sequestration and coastal protection but are especially vulnerable to extreme heat events. Elevated temperatures can disrupt seagrass photosynthesis and metabolic processes, leading to reduced growth and heightened susceptibility to disease (Deguette et al., 2022; Guerrero-Meseguer et al., 2020; Sawall et al., 2021; Winters et al., 2011). With repeated heatwave exposure and limited recovery periods, seagrass meadows may suffer severe declines, threatening their ability to deliver key ecosystem services such as carbon storage, sediment stabilization, and habitat provision (Mazdiyasni et al., 2019). The compounded impacts of atmospheric and marine heatwaves thus pose an existential threat to intertidal seagrass ecosystems, highlighting the urgent need for targeted climate adaptation measures to mitigate these escalating thermal stresses and preserve the resilience of these essential marine habitats (Russo and Domeisen, 2023; Stillman, 2019).

5.3 Seagrass resilience to heatwaves

Seagrass resilience to heatwaves is a complex and multifaceted issue shaped by species-specific traits, geographical location, and concurrent environmental stressors (Berger et al., 2024;

Canadell and Jackson, 2021; Hatum et al., 2024). As climate change drives the frequency and intensity of marine heatwaves, understanding these dynamics becomes crucial, especially for temperate seagrass meadows composed of slow-growing, long-lived species like *Posidonia*, *Amphibolis*, and *Zostera.* These species, unlike their tropical counterparts, tend to be highly vulnerable to abrupt environmental changes, struggling to recover from disturbances due to their slower growth and longer lifespan. In contrast, colonizing species typical of tropical regions, such as *Halodule*, *Halophila*, and *Syringodium*, demonstrate greater resilience to warming events and marine heatwaves due to their rapid growth and more flexible life strategies (O'Brien et al., 2018).

The impact of heatwaves on seagrass ecosystems is further intensified by tidal variations, especially in temperate regions with tidal ranges that can exceed 10 meters, such as Mont Saint Michel Bay in France. This pronounced tidal amplitude affects intertidal seagrasses like *Zostera noltei*, which experience varying durations of air exposure based on their position within the intertidal zone. During neap tides, seagrasses situated higher in the intertidal zone may remain exposed for extended periods, heightening their vulnerability to atmospheric heatwaves. Conversely, seagrasses in the lower intertidal zone experience prolonged immersion throughout the tidal cycle, making them susceptible to extreme marine heat events. As a result, both marine and atmospheric heatwaves can increase stress on seagrass meadows, with effects that may be intensified by tidal timing and amplitude. In our study, we observed a darkening effect in the seagrasses, despite our experimental setup simulating tides with consistent 6-hour exposure periods. In natural settings, however, bathymetry plays a significant role, and depending on depth variations, seagrasses may be exposed for even longer durations during low tides. This suggests that *in situ* exposure times could exacerbate the stress effects observed, as seagrasses may endure prolonged air exposure beyond what our controlled conditions have replicated.

Species like *Zostera noltei* display distinct seasonal patterns that become increasingly pronounced at higher latitudes (Davies et al., 2024a, 2024b). These patterns reflect the species' adaptation to seasonal variations in temperature and light, but they also make *Z. noltei* particularly sensitive to extreme events depending on their timing. For instance, if a heatwave coincides with early developmental stages or occurs after the biomass peak when leaf senescence has begun, the impact on meadow resilience can be severe.

Environmental conditions can moderate seagrass resilience to thermal stress. Seagrass meadows located in areas benefiting from tidal cooling or positioned further from the warmer edges of their geographical ranges often experience reduced heat stress, resulting in higher shoot densities and enhanced resilience (Berger et al., 2024; Canadell and Jackson, 2021). In these cooler areas, *Zostera noltei* exhibits high survival and photosynthetic performance up to 37°C, though temperatures above 39°C lead to near-total mortality within days, underscoring the species' sensitivity to temperature thresholds that may become increasingly common under climate change scenarios (Franssen et al., 2014). Moreover, the frequency, duration, and intervals between heatwaves significantly affect seagrass biomass and recovery; prolonged and frequent heatwaves reduce resilience and complicate recovery processes (Hatum et al., 2024).

The influence of other stressors, such as eutrophication and sulfide accumulation, complicates this resilience. Increased sediment sulfide levels, which often accompany nutrient enrichment, can be toxic to seagrasses, particularly under elevated temperatures that amplify sulfide toxicity. *Zostera noltei*, for example, has a mutualistic relationship with lucinid clams that helps detoxify sulfides. However, this interaction is compromised at high temperatures, reducing the efficacy of sulfide removal and further inhibiting nutrient uptake, growth, and overall resilience (De Fouw et al., 2022).

In a simulated heatwave experiment on *Zostera noltei*, resilience was evident under short-term moderate stress, with no significant changes in photosynthetic performance or survival (Franssen et al., 2014; Massa et al., 2009). However, prolonged or more intense heat events posed a greater challenge, highlighting the species' limited capacity to withstand chronic thermal stress. The long-term impact of heatwaves is especially evident in changes to seagrass cover (SC). Before the heatwave event on 1st September 2021, impacted seagrass patches exhibited higher SC than non-impacted areas. However, during the event on 6th September, SC in the impacted patch experienced a sharp decline. By 8th October, one month post-event, the SC of impacted seagrasses, initially higher, had fallen below that of non-impacted seagrasses (Figure 10). The observed decrease in seagrass cover on 6th September was primarily due to leaf darkening, which influenced the NDVI measurements used to estimate SC. This darkening effect reduced the NDVI values, creating an apparent decrease in SC in the impacted patch when, in reality, the density of seagrass remained stable (Figure 5). The bias introduced by remote sensing in this instance reflects a limitation in accurately capturing true seagrass cover during stress events. It was only after the heatwave that leaves began to detach, leading to an actual decline in seagrass density. This delayed physical response underscores how extreme events can compromise seagrass resilience, leaving previously robust patches in a weakened state compared to less disturbed areas (Figure 10).

On a physiological level, seagrasses possess coping mechanisms such as photoprotective responses and heat-responsive gene expression, including the activation of heat-shock proteins (HSPs) and other stress-related genes (Hughes and Stachowicz, 2004; Reusch et al., 2005). *Zostera noltei*, in particular, has shown differential gene expression responses to heat stress, with a variety of genes involved in protein folding, membrane stability, and reactive oxygen species scavenging playing critical roles. However, the rapid pace of climate change raises concerns about whether these adaptations can keep up with the increasing frequency and severity of thermal events (Franssen et al., 2014).

5.4 Big picture

Given the combined pressures of temperature extremes, eutrophication, and other anthropogenic impacts, targeted management strategies are essential for enhancing seagrass resilience (Loarie et al., 2009). Approaches such as reducing local stressors, cultivating heat-tolerant genotypes, and investing in restoration initiatives are vital to supporting these ecosystems in a warming climate. Although challenges remain, the adaptability and potential resilience of certain seagrass species offer hope for their persistence amid accelerating ecological shifts. In particular, regions that can buffer seagrasses from extreme stressors or provide cooler refuges may play a critical role in maintaining these valuable ecosystems in the face of global climate change (Canadell and Jackson, 2021; De Fouw et al., 2022).

Seagrass meadows function as foundational components of coastal ecosystems, sustaining diverse marine communities by providing essential habitats, nursery grounds, and trophic resources for fish, invertebrates, and migratory birds (Zoffoli et al., 2023). Their dense canopies stabilize sediment and protect shorelines from erosion, an increasingly crucial role as sea levels rise due to climate change (Folmer et al., 2012; Gacia et al., 1999). Recurrent heatwave events, which induce physiological stress like leaf darkening, can severely diminish seagrass density, thereby reducing their effectiveness in sediment stabilization and wave attenuation, ultimately increasing the risk of coastal erosion (Calleja et al., 2007). From a biodiversity perspective, degradation of these meadows disrupts intricate food webs, impacting commercially significant fish and shellfish populations that rely on seagrass for sustenance and refuge. This loss can reduce local fisheries' productivity and threaten the livelihoods of coastal communities (Unsworth and Cullen-Unsworth, 2014). Furthermore, as seagrasses decline, their capacity to act as a blue carbon sink—critical for climate mitigation—also diminishes, inadvertently contributing to increased atmospheric carbon levels (Armitage and Fourqurean, 2016; Samper-Villarreal et al., 2020)

6 Conclusion

This research has investigated the effects of both marine and atmospheric heatwaves on the intertidal seagrass *Zostera noltei*, a critical component of coastal ecosystems facing increased thermal stress due to climate change. By examining reflectance and pigment composition under controlled experimental conditions and validating these findings with satellite data, we aimed to understand how extreme heat events impact seagrass health and assess the potential of remote sensing to monitor these effects effectively. Our findings reveal that heatwaves lead to substantial declines in seagrass reflectance, particularly in the green and near-infrared regions, likely driven by pigment degradation and structural damage. This change is reflected in significant reductions in key vegetation indices such as NDVI and GLI. The Seagrass Darkening Index (SDI), developed in this study, successfully detected seagrass darkening, a visible symptom of heatwave stress, demonstrating the viability of spectral monitoring to capture early-stage impacts of heat events on intertidal ecosystems. By connecting our findings with satellite data, we have also confirmed the broader spatial impact of heatwaves on seagrass meadows in Quiberon, France. The correlation between heatwave exposure and darkening of seagrass suggests that remote sensing, combined with targeted field observations, can enhance our understanding of ecosystem responses to climate-driven thermal events. These results advocate for integrating continuous spectral monitoring into conservation strategies, as it can help predict the resilience of these ecosystems and guide adaptive management practices. As climate change accelerates, with an anticipated increase in the frequency and intensity of heatwaves, the vulnerability of intertidal zones to such events will likely intensify. This study not only underscores the importance of seagrass meadows in coastal ecosystems but also highlights the urgency of protecting these critical habitats. Future work should focus on refining remote sensing tools and examining the cumulative effects of repeated heatwave events to support the resilience and conservation of intertidal seagrass meadows in a warming world.

7 Annexes

7.1 Annexes A - Temperatures of the experiment

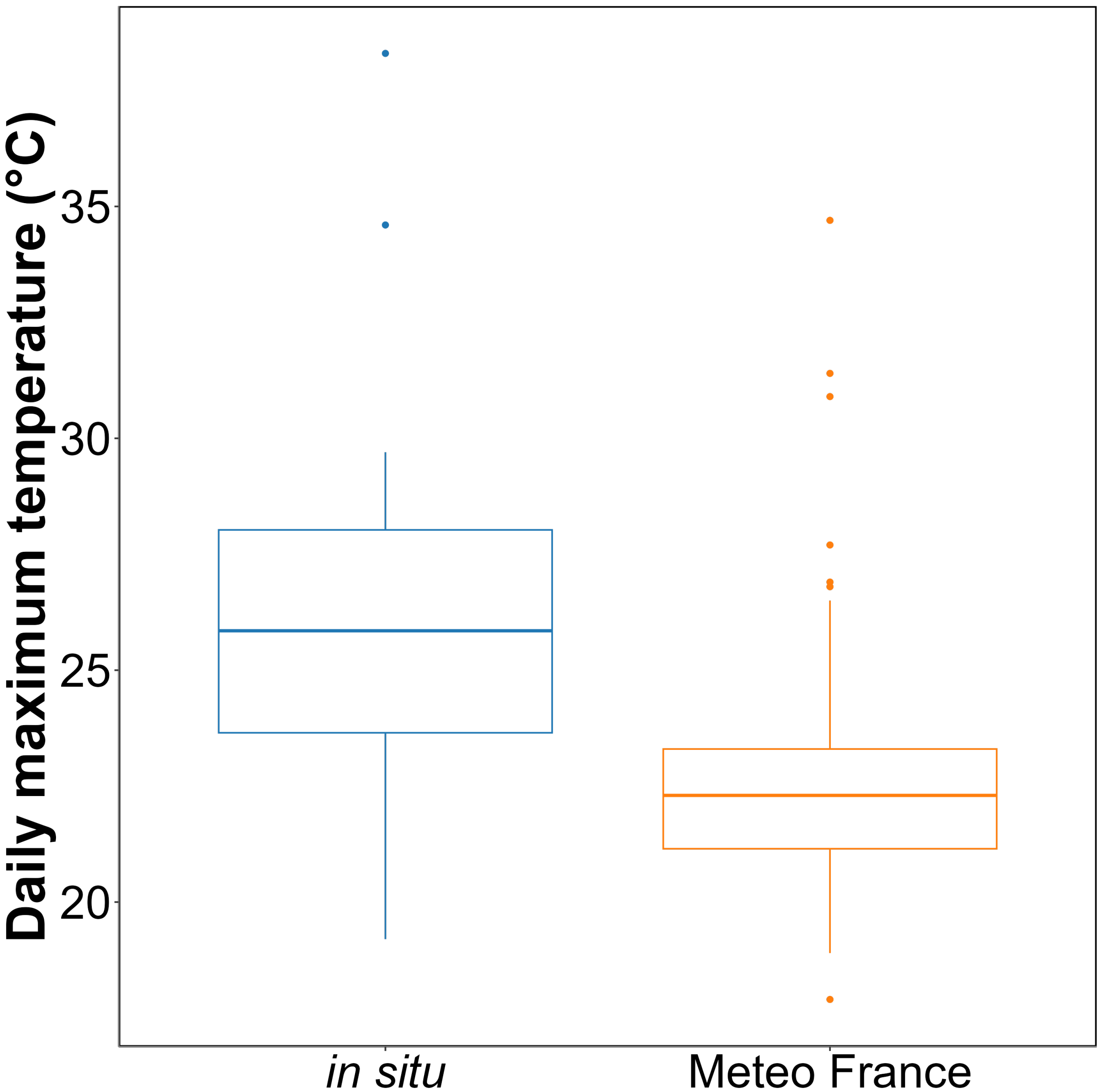

Figure 13: Annexe A1: Comparison of daily maximum temperatures in August measured using an in-situ sensor (blue) and retrieved from Meteo France (orange). The solid line in the middle of the boxplot represents the median, the two ends of the box represent the 25th and 75th percentiles, and the whiskers represent values that are no more than 1.5 times the interquartile range.

On average, *in situ* temperatures were 3 ± 3.2°C higher than those recorded by Meteo France. Additionally, temperatures recorded by Meteo France were more stable than those from the *in situ* sensors, likely due to the sheltered and shaded location of the Meteo France equipment. This difference was used to adjust heatwave temperatures measured by Meteo France to better reflect the conditions experienced by the seagrasses.

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